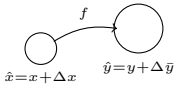
Taylor (extra, if a problem needs one)
backward: $f(x-h)=f(x)-f'(x)h+\frac{1}{2}f''(x)h^2-\frac{1}{3!}f'''(x)h^3+\cdots$
vector: $f(\mathbf{x}+\Delta\mathbf{x})=f(\mathbf{x})+\nabla f\,\Delta\mathbf{x}+\frac{1}{2}\Delta\mathbf{x}^T H\,\Delta\mathbf{x}+\cdots$
centered from Taylor: fwd−bwd $\Rightarrow f'(x)=\frac{f(x+h)-f(x-h)}{2h}+O(h^2)$

L2 System Models

Std linear form $\dot{\mathbf{x}}=A\mathbf{x}+B\mathbf{u}$, $\mathbf{y}=C\mathbf{x}+D\mathbf{u}$; nonlinear form $\dot{\mathbf{x}}=f(\mathbf{x},\mathbf{u})$, $\mathbf{y}=h(\mathbf{x},\mathbf{u})$.
Example: ODE to state (car suspension). $m\ddot{y}+c\dot{y}+ky=u$; choose $\mathbf{x}=[y,\dot{y}]^T$ (stack the derivatives):
$A=\begin{bmatrix}0 & 1\\ -k/m & -c/m\end{bmatrix}$, $\mathbf{b}=\begin{bmatrix}0\\ 1/m\end{bmatrix}$.
Homog $\mathbf{u}=0$: $\mathbf{x}(t)=e^{At}\mathbf{x}(0)=\Phi(t)\mathbf{x}(0)$
Matrix exp $e^{At}=I+At+\frac{1}{2}(At)^2+\cdots$: A nilpotent ($A^n=0$): series stops; else diag $A=V\Lambda V^{-1}$, $e^{At}=Ve^{\Lambda t}V^{-1}$, $e^{\Lambda t}=\mathrm{diag}(e^{\lambda_i t})$.
Discrete model $\mathbf{x}_{k+1}=A_d\mathbf{x}_k+B_d\mathbf{u}_k$; $A_d=e^{Ah}$, $B_d=\int_0^h e^{A\tau'}d\tau'\,B$, $f_s=1/h$.
Get A_d, B_d $\begin{bmatrix}A_d & B_d\\ 0 & I\end{bmatrix}=\exp\left(\begin{bmatrix}A & B\\ 0 & 0\end{bmatrix}h\right)$; stops if A nilpotent.
Example: double integrator (discrete). $\ddot{p}=u$:
$A=\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}$, $\mathbf{b}=\begin{bmatrix}0\\ 1\end{bmatrix}$, $A^2=0$, $A_d=\begin{bmatrix}1 & h\\ 0 & 1\end{bmatrix}$, $\mathbf{b}_d=\begin{bmatrix}h^2/2\\ h\end{bmatrix}$,
$p_{k+1}=p_k+hv_k+\frac{1}{2}h^2a_k$, $v_{k+1}=v_k+ha_k$.
Example: single integrator (discrete). $\dot{x}=u$, $y=x$, zero-order hold, period h. Over $[t_k,t]$: $x(t)=x(t_k)+\int_{t_k}^{t}u(\tau)\,d\tau=x(t_k)+(t-t_k)\,u(t_k)$ (ZOH: u held constant $=u_k$). At $t=t_{k+1}$ (so $t-t_k=h$): $x_{k+1}=x_k+h\,u_k$, $y_k=x_k$.

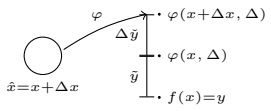
L3 Conditioning & Stability



Errors abs $|\Delta x|,|\Delta y|$; rel $\delta x=\frac{|\Delta x|}{x}$, $\delta y=\frac{|\Delta y|}{y}$.
Condition number (problem sensitivity). Taylor $f(x+\Delta x)=f(x)+f'(x)\Delta x+O(\Delta x^2)\Rightarrow\Delta\bar{y}=f'(x)\Delta x$. Abs $\mathrm{Cond}_x\approx|f'(x)|$ (want small); rel $\mathrm{cond}_x\approx|K_x|$, $K_x=f'(x)\frac{x}{f(x)}$ (want <1).
Example: Condition number.

- linear $y=ax$: $K_x=1$ (percentage error unchanged).
- differential eq $\dot{y}=\lambda y$: $\frac{\Delta\bar{y}}{\Delta\lambda}=y_0te^{\lambda t}$, $\lim_{t\to\infty}\left|\frac{\Delta\bar{y}}{\Delta\lambda}\right|=0$ ($\lambda<0$, well cond) or ∞ ($\lambda\geq0$, ill cond): asymptotically stable ODEs are well conditioned.
- evaluate $g(y)$: Cond$\approx|g'(y)|$ (g steep: ill cond; g flat: well cond).
- root finding $g(y)=0$: Cond$\approx\frac{1}{|g'(y)|}$, the reciprocal of evaluating g (inversely proportional to slope; steep g: sharp root but error-amplifying evaluation).

Approx algorithm $\varphi(x,\Delta)$ approximates f; $\Delta=$ step.



$\Delta\check{y}=\underbrace{\varphi(x+\Delta x,\Delta)-\varphi(x,\Delta)}_{\Delta\check{y}}+\underbrace{\varphi(x,\Delta)-f(x)}_{\bar{y}}$; $\bar{y}$ approx error, $\Delta\check{y}$ propagated input error, $\Delta\bar{y}$ total. Propagated by Taylor $\Delta\check{y}\approx\frac{d\varphi}{dx}\Delta x$.
Stability (algorithm, not problem): $\left|\frac{\Delta\check{y}}{\Delta x}\right|=\left|\frac{d\varphi}{dx}\right|$: <1 stable, >1 unstable, $=1$ marginal.
Example: Numerical Instability. $\dot{y}=-\alpha y$ ($\alpha>0$, well conditioned). Analytic: $y(t)=y_0e^{-\alpha t}$.

- Forward Euler: $y(t+h)=(1-\alpha h)y(t)\Rightarrow\varphi(y)=(1-\alpha h)y$; $\left|\frac{d\varphi}{dy}\right|=|1-\alpha h|\leq1\Rightarrow0\leq h\leq2/\alpha$.
- Backward Euler: $\frac{y(t+h)-y(t)}{h}=-\alpha y(t+h)\Rightarrow y(t+h)=\frac{y(t)}{1+\alpha h}\Rightarrow\varphi(y)=\frac{y}{1+\alpha h}$; $\left|\frac{d\varphi}{dy}\right|=\frac{1}{1+\alpha h}<1$ for all $h>0$.

Same problem, swap the algorithm: instability is a property of the algorithm, not the problem.
Consistency φ consistent if $\tilde{y}=\varphi(x,\Delta)-f(x)\to0$ as $\Delta\to0$, i.e. $\lim_{\Delta\to0}\varphi(x,\Delta)=f(x)$.
Example: Consistency. Two more derivatives; consistent, what order? (L'Hôpital: both are $\frac{0}{0}$ at $\Delta\to0$, differentiate top & bottom in Δ):

- backward $\frac{f(x)-f(x-\Delta)}{\Delta}\xrightarrow{\text{L'H}}\frac{f'(x-\Delta)}{1}\to f'(x)$: consistent.
- centered $\frac{f(x+\Delta)-f(x-\Delta)}{2\Delta}\xrightarrow{\text{L'H}}\frac{f'(x+\Delta)+f'(x-\Delta)}{2}\to f'(x)$: consistent.

Order by Taylor: backward $=f'(x)-\frac{1}{2}f''(x)\Delta\Rightarrow p=1$; centered $=f'(x)+\frac{1}{6}f'''(x)\Delta^2\Rightarrow p=2$.

L4 Root Finding

Problem find $x^\star$ with $f(x^\star)=0$.
IVT $f:[a,b]\mapsto\mathbb{R}$ continuous, $f(a)<u<f(b)$ or $f(b)<u<f(a)\Rightarrow$ exists $z\in(a,b)$ with $f(z)=u$. (root: $u=0$.)
Smoothness C^0 continuous; C^1 continuous f'; C^2 continuous f''. $|x|$ is C^0 not C^1; x^2 is C^∞.
Convergence $E^{(k)}=|x^{(k)}-x^\star|$: linear $E^{(k+1)}=cE^{(k)}$ ($c<1$); superlinear $1<p<2$; quadratic $p=2$ (digits double).
Bisection iters to tol ε: $n\geq\log_2\frac{b-a}{\varepsilon}$. Fixed point converges iff $|g'(x^\star)|<1$.
Lipschitz g is Lipschitz (constant c) if $|g(x)-g(y)|\leq c|x-y|$ for all x,y (c bounds the slope). A contraction is Lipschitz with $c<1$; then g pulls points closer, so fixed-point iteration converges.
Methods (update; requirement; convergence; note)

- bisect: $m=\frac{a+b}{2}$, keep half where f flips sign; f continuous (C^0) + bracket $f(a)f(b)<0$; linear $c=\frac{1}{2}$; always converges, slow
- fixed pt: $x_{k+1}=g(x_k)$, $f=0\Leftrightarrow x=g(x)$; g a contraction: $|g(x)-g(y)|\leq c|x-y|$, $c<1$ (i.e. $|g'|\leq c<1$); linear; fails if $c\geq1$
- Newton: $x_{k+1}=x_k-\frac{f(x_k)}{f'(x_k)}$; $f\in C^1$, $f'(x^\star)\neq0$; evaluates f'; quadratic $p=2$; fastest
- secant: $x_{k+1}=x_k-f_k\frac{x_k-x_{k-1}}{f_k-f_{k-1}}$; $f\in C^1$ (f' exists) but f' never evaluated (slope from 2 pts); superlinear $p\approx1.62$; 2 starts
- systems: $\mathbf{x}_{k+1}=\mathbf{x}_k-J^{-1}\mathbf{f}(\mathbf{x}_k)$, J Jacobian

Example: bisection. $x+\cos x$ on $[-5,5]$: $f(-5)<0$, $f(5)>0$; $m=0$: $f>0\to(-5,0)$; $m=-2.5$: $f<0\to(-2.5,0)$.

L5 Numerical Integration & Differentiation

Numerical integration (quadrature) approximate the area $\int_a^b f(x)\,dx$ using a few function values $(x_i,f(x_i))$; more points means more accurate but slower. Over one interval $[x_i,x_{i+1}]$ with width $\Delta=x_{i+1}-x_i$ and mid$=\frac{x_i+x_{i+1}}{2}$:

- midpoint: $\int_{x_i}^{x_{i+1}}f(x)\,dx\approx\Delta\,f(\text{mid})$, error $O(\Delta^2)$
- trapezoid: $\int_{x_i}^{x_{i+1}}f(x)\,dx\approx\Delta\,\frac{f(x_i)+f(x_{i+1})}{2}$, error $O(\Delta^2)$
- Simpson: $\int_{x_i}^{x_{i+1}}f(x)\,dx\approx\Delta\,\frac{f(x_i)+4f(\text{mid})+f(x_{i+1})}{6}$, error $O(\Delta^4)$

Numerical differentiation approximate the slope $f'(x)$ from nearby values, step Δ:

- forward: $f'(x)\approx\frac{f(x+\Delta)-f(x)}{\Delta}$, error $O(\Delta)$
- backward: $f'(x)\approx\frac{f(x)-f(x-\Delta)}{\Delta}$, error $O(\Delta)$
- centered: $f'(x)\approx\frac{f(x+\Delta)-f(x-\Delta)}{2\Delta}$, error $O(\Delta^2)$

choosing Δ: too big gives a poor approximation; too small gives round-off noise.
Algorithm order the algorithm error $\tilde{y}=\varphi(x,\Delta)-f(x)\propto\Delta^p$, i.e. $O(\Delta^p)$; p is the order (e.g. centered diff $p=2$).
Consistency (L5 form) consistent if $\lim_{\Delta\to0}\varphi(x,\Delta)=f(x)$.
Diff stability an input error grows by $|f''(x)|$; stable iff $|f''(x)|<1$ (fwd & centered). f'' is the conditioning of differentiation itself, so no formula fixes a bad one.
Integrate vs solve ODE $\int f\,dt$ open loop; $\dot{x}=f(x,t)$ feeds back x, so stability matters.
Existence/uniqueness (theorem) if f is continuous and Lipschitz, $\|f(\mathbf{u})-f(\mathbf{v})\|_2\leq L\|\mathbf{u}-\mathbf{v}\|_2$ for some $L\geq0$, then $\dot{\mathbf{x}}=f(\mathbf{x})$ has exactly one solution for all $t\geq0$, regardless of initial conditions.
Time-varying→invariant augment $g=t$: $\frac{d}{dt}[\mathbf{x};g]=[f(\mathbf{x},g);1]$.
Example: Linear ODE. $\dot{y}=-\alpha y$. Separable: $\frac{dy}{y}=-\alpha\,dt$, integrate $\ln y=-\alpha t+C\Rightarrow y(t)=y_0e^{-\alpha t}$. Exact one-step map $y(t+h)=e^{-\alpha h}y(t)$; expand $e^{-\alpha h}=1-\alpha h+\frac{1}{2}\alpha^2h^2-\cdots$: $y(t+h)=(1-\alpha h+\frac{1}{2}\alpha^2h^2-\cdots)\,y(t)$. Keep the 1st order term $\Rightarrow y(t+h)\approx(1-\alpha h)y(t)=$ forward Euler.

L6 ODE Solvers I (Euler)

Test equation $\dot{x}=\lambda x$: unstable $\lambda>0$, stable $\lambda=0$, asympt stable $\lambda<0$. Problem cond $\left|\frac{\Delta\bar{x}}{\Delta x_0}\right|=e^{\lambda T}$: ill cond $\lambda>0$, well cond $\lambda<0$.
Analyze with $\lambda<0$. (well conditioned $\neq$ numerically stable.)
Forward Euler explicit (uses past x_k, f_k); a time stepping scheme. Approx derivative $\dot{x}_k=f_k=\frac{x_{k+1}-x_k}{h}+O(h)$, rearrange $\Rightarrow x_{k+1}=x_k+hf_k$.
LTE by Taylor:
$\dot{x}(t_{k+1})=x(t_k)+h\dot{x}(t_k)+\frac{h^2}{2}\ddot{x}(t_k)+O(h^3)=x_k+hf_k+\underbrace{O(h^2)}_{\text{LTE}}$ so LTE $O(h^2)$.
GTE $=O(\frac{1}{h})\cdot O(h^2)=O(h)$ over $K=T/h$ steps $\Rightarrow$ first order.
Stable iff $|1+h\lambda|<1$ (disk centre -1, radius 1 in $h\lambda$); real $\lambda<0$: $0<h<2/|\lambda|$.
Backward Euler implicit (f_{k+1}, so x_{k+1} on both sides): $x_{k+1}=x_k+hf_{k+1}$. Implement: invert f, root-find, or approximate.
Test eq $x_{k+1}=\frac{x_k}{1-h\lambda}$, stable iff $\left|\frac{1}{1-h\lambda}\right|<1$ (outside disk centre $+1$; stable for any h when $\lambda<0$).
LTE vs GTE LTE one step; GTE over $K=O(1/h)$ steps, one order lower; GTE is the method order.
Stiff $|\lambda_{\max}|/|\lambda_{\min}|$ large. Explicit h set by fastest mode (largest $|\lambda|$) $\Rightarrow$ tiny h; backward Euler handles it.
Matrix $\dot{\mathbf{x}}=A\mathbf{x}$ test each eigenvalue λ_i of A; need every $h\lambda_i$ in the region.
Example: Stable or Not? $\ddot{x}+2\dot{x}+2x=0$: char. eq $\lambda^2+2\lambda+2=0\Rightarrow\lambda=\frac{-2\pm\sqrt{4-8}}{2}=-1\pm i$. Forward Euler always tests $|1+h\lambda|<1$ on each eigenvalue; $h=0.5$: $|1+0.5(-1\pm i)|=|0.5\pm0.5i|=\sqrt{0.5^2+0.5^2}=\frac{1}{\sqrt{2}}<1$ stable.
Example: Stability of numerical differentiation (centered). $\varphi(x,\Delta)=\frac{f(x+\Delta)-f(x-\Delta)}{2\Delta}$. Perturb the input $x\to x+\Delta x$ and Taylor each term:
$\Delta\check{y}=\varphi(x+\Delta x,\Delta)-\varphi(x,\Delta)=\frac{\Delta x\,f'(x+\Delta)-\Delta x\,f'(x-\Delta)}{2\Delta}+O(\Delta x^2)=(f''(x)+O($
So $\left|\frac{\Delta\check{y}}{\Delta x}\right|=|f''(x)|$, stable iff <1.